

Me Re Diam

Simulating socio-agricultural practices of agro-pastoralists in Sénégal

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DSCATT studies agricultural strategies to enforce carbon sequestration (4 ‰)

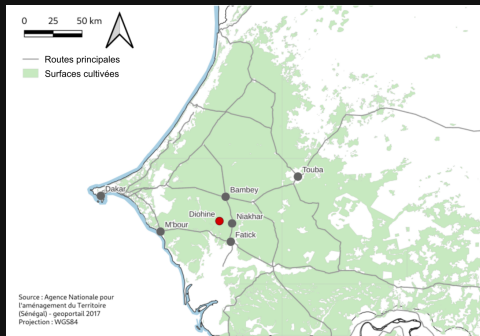
- different countries : Sénégal, Zimbabwe, France,
- various scales : parcel, farm, village / small region
- IRD , CIRAD , INRAE , AGROOF , ISRA , Univ. of Zimbabwe 

WP3 : Socio economic, policy and environmental levers for long-term soil C sequestration at the territory level

→3.2 Social, institutional, and economic context potentially driving C fluxes and sequestration

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Target system : Sassem district
food sustainability & community
fallow preservation



Groundnut basin

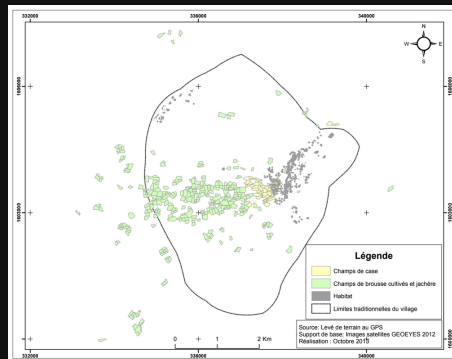
Sub-sahelian climate

Diohine village (red dot), in the Fatick region

Sassem : district of farmers, agro-pastors, cattle breeders

488 inhabitants in 2020

180 ha of arable lands



Sassem parcels in 2013, bushfields in green

From the XVIIth to the 70's:

- Fallow land and savannas shrink, selective deforestation
- Forced transhumance → lesser manure input
- Rapid population growth

Nowadays :

- Fallow almost disappeared
- 50% decrease in yields since the 70s (Pieri 1989)
- Limits of arable land reached long ago
- fragility of soil **fertility** and agricultural systems (Faye et al. 2021)

But !

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But !

Scarcity fosters **mutual aid and solidarity** (Delay and Piou 2019)

Diohine community fallow land **has managed to persist**

What are the conditions for maintaining community fallow land in Diohine ?

What are the conditions for sustaining village livelihood at both village and household scale ?

The *Me Re Diam* model ("I see you" in sereer)

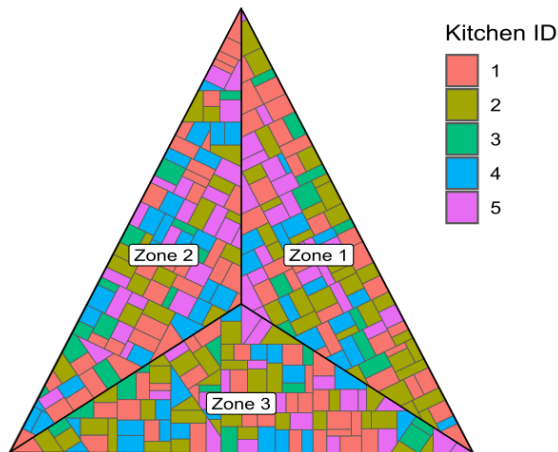
Agents are **kitchens** ($\approx$ households)

- Kitchens possess some **parcels** and **cattle** (standard TLU cow)
- Kitchens cattle is aggregated in the village's **herd** $\approx$ 85 TLU
- Kitchen cultivate their parcels according to the number of mouth to feed (500g/person/day) and the available land

Mostly 3-years crop rotation : Millet $\rightarrow$ Peanut $\rightarrow$ Fallow

In case of insufficient land or food : **solidarity** manifests by **parcel loans** and **food donations**.

Synthetic parcels : 5 kitchens, Gini index of 0.2



Controlable settings :

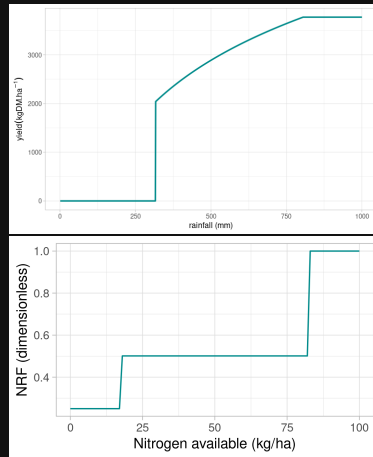
- number of kitchens
- inequity in kitchens owned surface and number of parcels
- crop rotation zoning (2-years or 3-years)

Thanks to the work of parcel recomposition by Maxime Colomb, [LASTIG/INRIA](#)

Plants growth is limited by **water** and available **Nitrogen**

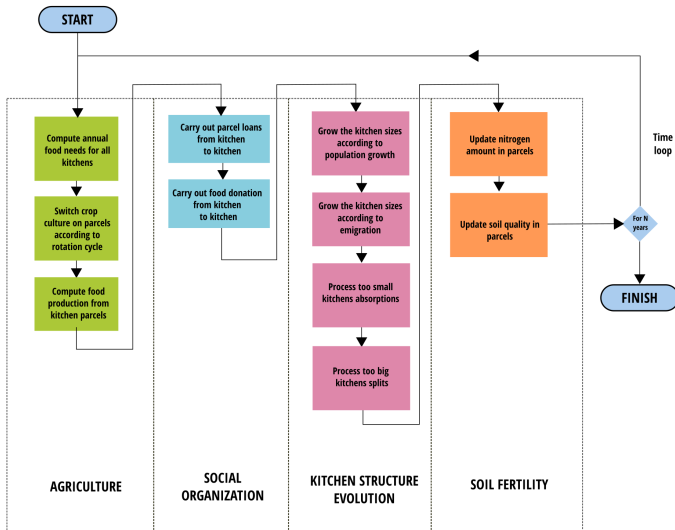
- **manure** (cow dungs) increase available N
- **faidherbia trees** increase available N

Millet, Peanut and Fallow have their own equations and residues ratios



Yield equations and N mechanics taken from (Grillot et al. 2018) after a simulation model calibrated on empirical Sahelian data : CELSIUS (Ricome et al. 2017)

- Typical kitchen size is 16
- Village's population increases by natural growth rate (Bernouilli process)
- Insufficient food (after solidarity) causes emigration in nearby cities
- Too big kitchens split: land and herd are split in proportion
- Too tiny kitchens are absorbed by a bigger one: land and herds are merged



Fallow biomass feeds the village's herd, manure benefits partly to everyone

Fallow is kept untouched **if possible** In case of land shortage,

- a kitchen may cultivate on its own fallow parcels
- a kitchen may loan its fallow parcel to another kitchen in need

but only as a **last resort** (Roach P. 2000)

Calibration experiment

Demographics census data from 1995 to 2020

(Delaunay, Desclaux, and Sokhna 2018)

Growth rate : 1.31%

Rainfall data, from the CHELSA open dataset (Karger et al. 2018; Karger et al. 2016)

Agricultural ground truth from actors:

Millet yield : 700kg.ha^{-1} (grain)

Groundnut yield : 400kg.ha^{-1} (nuts and shells)

Fallow preservation data measured by (Odru 2013)

60% intact fallow in 2012, 40% in 2020

Sassem population from 1995 to 2020



Diohine rainfall from 1995 to 2020



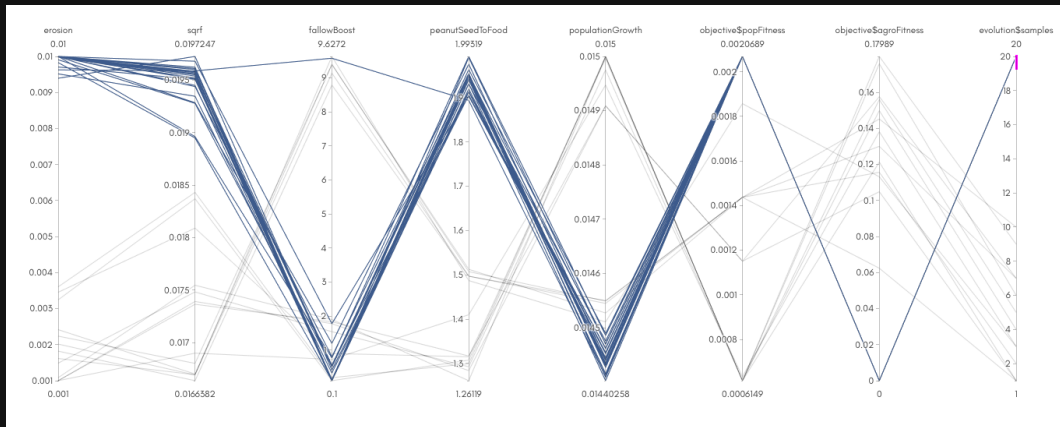
Criteria

- #1 population dynamics close to observed demographic data
- #2 an agricultural system close to groundtruth :
 - Millet crop yields around 700 kg.ha^{-1}
 - intact fallow ratios around 60% in 2012 and 40% in 2020

Criteria may be conflicting : solutions are not unique (Pareto Front sometimes)



Here : lightning fast model Scala implementation (1s/run) allow calibration in $< 1\text{h}$ on a 8-cores laptop

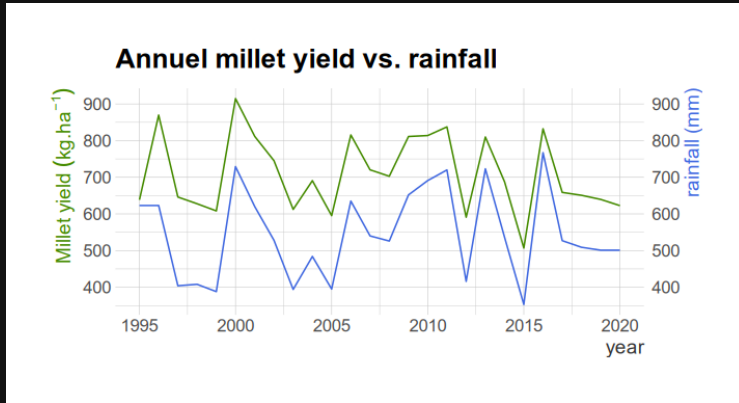


Good news : fitness values very low and stable replications match !

"Solutions" are the values of the first 5 technical parameters

Height dynamics to assess a simulation:

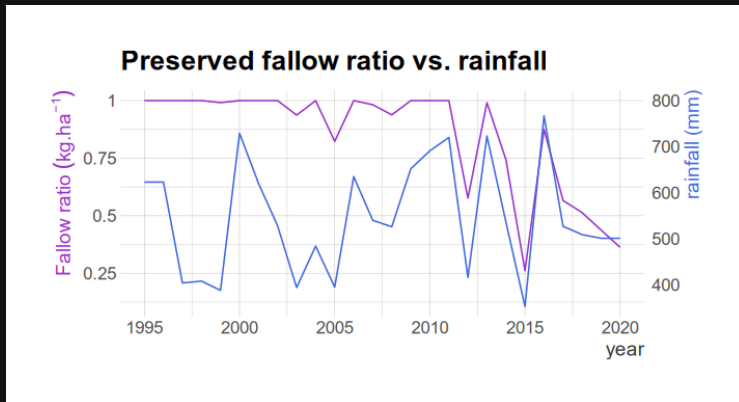
- the population dynamics
- the **intact fallow area dynamics**
- the emigration dynamics (total)
- the village herd size dynamics,
- the annual soil quality (median over parcels)
- the **millet yields dynamics** (averaged over kitchens)
- the cumulative loaned area
- the annual kitchen food stress (averaged over kitchens)



N.B. Usually forbidden double Y-axis plot, shown here with luckily compatibles ranges

Rainfall drives the Millet Yield (same for peanut)

Other practices to increase yield are **marginally efficient** ? Available N mechanism useless ?



N.B. Usually forbidden double Y-axis plot, shown here with fake ticks labels

Fallow is like a "buffer" : intact until rains is missing or population overcoming parcels production capacity

Fallow **compensates** insufficient land or rainfall against population growth **temporarily**

After Calibration findings

Conclusion : The answer(s)

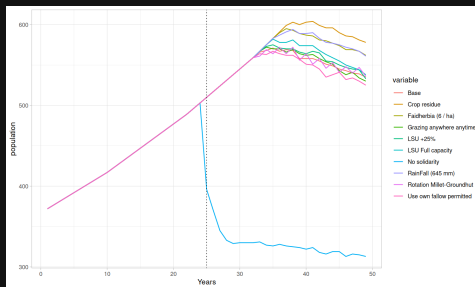
According to the model (calibration + single bifurcations ceteris paribus)

What are the *conditions for sustaining village livelihood at both village and household scale* ?

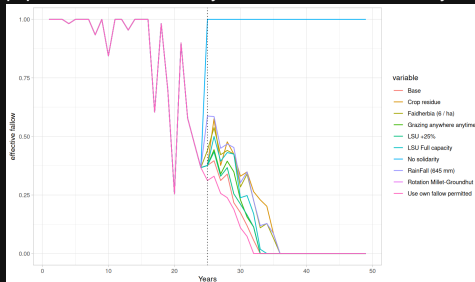
- Solidarity (loans and donations) **mandatory**
- Faidherbiae helps
- Mulching is an interesting option

What are the *conditions for maintaining community fallow land in Diohine* ?

- no "plausible" solutions, only extreme
- remove solidarity (and accept -35% population) **unthinkable in Sassem**
- Population growth severe mitigation ?



population bifurcation. Sky-blue line is selfish kitchen dynamic



Intact fallow bifurcation. Sky-blue line is selfish kitchen dynamic

Model to predict **anticipate** and **discuss**

Interplay of alternative mechanisms is not (yet) explored.

Model scope adequation / completeness ?

Fieldwork side

- (i) **Actors emphasis on social peace and daily solidarity**
- (ii) Natality low-key reduction (from 5 to 3 kids by kitchen)
- (iii) Some young adults prefer to work in surrounding cities, sending money to their kitchen

Model side

- (i) **System collapse with selfish kitchens**
- (ii) Increasing population is bound to shrink the fallow, no feedback loop on population growth
- (iii) No food donation from remote kitchen members

DSCATT project <https://dscatt.net>

Code repository <https://github.com/ElCep/DSCATT/tree/fertility>

Openmole website next.openmole.org

References



Delaunay, Valérie, Alice Desclaux, and Cheikh Sokhna (2018). *Niakhar, mémoires et perspectives: recherches pluridisciplinaires sur le changement en Afrique*. fre. Dakar Marseille: l'Harmattan Sénégal IRD. isbn: 978-2-343-15671-2 978-2-7099-2669-0.



Delay, Etienne and Cyril Piou (Dec. 2019). "Mutual aid: When does resource scarcity favour group cooperation?" In: *Ecological Complexity* 40, p. 100790. issn: 1476-945X. doi: [10.1016/j.ecocom.2019.100790](https://doi.org/10.1016/j.ecocom.2019.100790). url: <http://www.sciencedirect.com/science/article/pii/S1476945X18300114> (visited on 10/15/2019).



Faye, Amy et al., eds. (2021). *Senegal - Land, climate, energy, agriculture and development: A study in the Sudano-Sahel Initiative for Regional Development, Jobs, and Food Security*. eng. ZEF Working Paper Series No. 202. doi: [10.22004/ag.econ.308808](https://doi.org/10.22004/ag.econ.308808).



Grillot, Myriam et al. (Sept. 2018). "Multi-level analysis of nutrient cycling within agro-sylvo-pastoral landscapes in West Africa using an agent-based model". In: *Environmental Modelling & Software* 107, pp. 267–280. issn: 1364-8152. doi: [10.1016/j.envsoft.2018.05.003](https://doi.org/10.1016/j.envsoft.2018.05.003). url: <https://www.sciencedirect.com/science/article/pii/S1364815217304000> (visited on 07/11/2024).



Karger, Dirk Nikolaus et al. (July 2016). "Climatologies at high resolution for the Earth land surface areas". In: *Scientific Data* 4.



Karger, Dirk Nikolaus et al. (2018). *Data from: Climatologies at high resolution for the earth's land surface areas*. doi: <https://doi.org/10.5061/dryad.kd1d4>.



Odru, Mariana (2013). "Flux de biomasse et renouvellement de la fertilité des sols à l'échelle du terroir. Etude de cas d'un terroir villageois sereer au Sénégal.". Master 2. Cergy-pontoise: ISTOM. (Visited on 06/03/2024).



Pieri, Christian (1989). "Fertilité des terres de savanes. Bilan de trente ans de recherche et de développement agricoles au Sud du Sahara". In: url: <https://api.semanticscholar.org/CorpusID:155370600>.



Reuillon, Romain, Mathieu Leclaire, and Sebastien Rey-Coyrehourcq (Oct. 2013). "OpenMOLE, a workflow engine specifically tailored for the distributed exploration of simulation models". Anglais. In: *Future Generation Computer Systems* 29.8, pp. 1981–1990. doi: [10.1016/j.future.2013.05.003](https://doi.org/10.1016/j.future.2013.05.003). url: <http://hal.archives-ouvertes.fr/hal-00840744>.



Ricome, Aymeric et al. (2017). "Are subsidies to weather-index insurance the best use of public funds? A bio-economic farm model applied to the Senegalese groundnut basin". In: *Agricultural Systems* 156, pp. 149–176. issn: 0308-521X. doi: <https://doi.org/10.1016/j.agsy.2017.05.015>. url: <https://www.sciencedirect.com/science/article/pii/S0308521X16306849>.

Thank You



Le marigot de Diohine, 2021



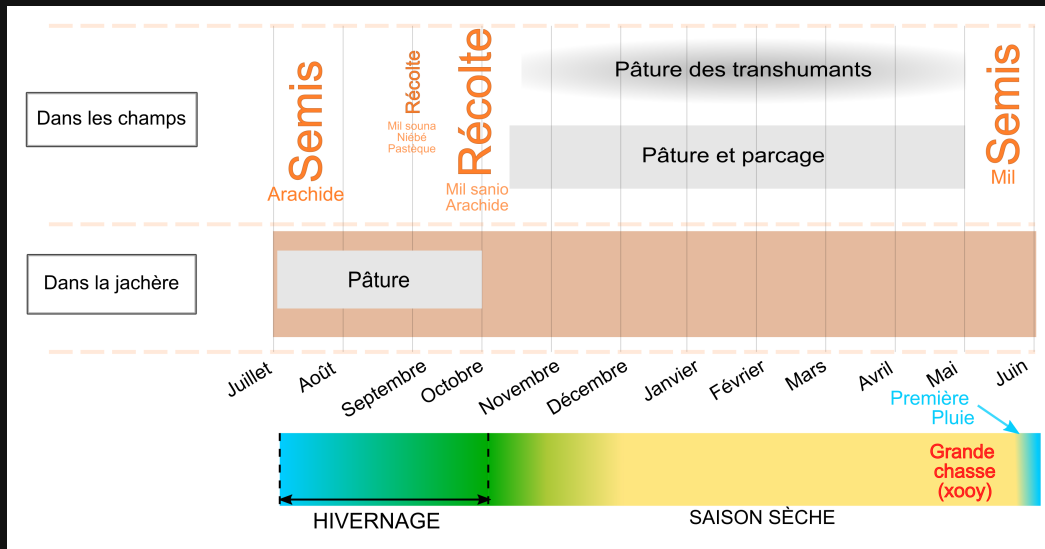
OpenMOLE
the model exploration software

 Taylored design of experiments platform to answer some questions : Sensitivity Analysis, Calibration , Diversity Search, etc.. (Reuillon, Leclaire, and Rey-Coyrehourcq 2013)

 Model agnosticism : model as a black box

 OpenMOLE handles jobs (runs) scheduling on HPC environments (or CPU cores).

Here : lightning fast model Scala implementation (1s/run) allow calibration in < 1h on a 8-cores laptop



calendrier "agronomique" en vigueur à Diohine

Engineering resilience : the ability of a system to return to an equilibrium state after a temporary perturbation by measuring how quickly a system's equilibrium is restored (Pimm 1984, Holling 1996).

Socio-technical resilience i.e. common points in a lot of resilience definitions :

- **Adaptability** and **flexibility**
- The ability to **self-organize**
- The ability to build and increase the capacity for learning and adaptation
- The ability to maintain links with other systems

Year	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007
Population	358	371	384	378	368	370	363	386	408	409	411	411	402
Year	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Population	417	434	455	450	460	477	468	467	466	475	487	492	488

Demographic evolution in Sassem (Delaunay, Desclaux, and Sokhna 2018)

Three "distances"

- simulated population close to final data population in 2020
- ajustement between simulated dynamics and observed dynamics (R^2)
- similar slope in simulated population dynamics vs data population over time ()

normalized to a score $\in [0; 1]$

Three "distances"

- preserved fallow proportion close to 60% in 2012
- preserved fallow proportion close to 40% in 2020
- millet yield $\approx 700kg.ha^{-1}$

normalized to a score $\in [0; 1]$

$$\begin{aligned} popFit = (&|5.67 - populationSlope| + \\ &|488 - lastPopulation| + \\ &|1 - populationRSquare|)/3 \end{aligned} \quad (1)$$

$$\begin{aligned} agroFit = &\text{if } 0.55 \leq ef_{2012} \leq 0.65 \text{ then } 0 \text{ else } |0.6 - ef_{2012}| + \\ &\text{if } 0.35 \leq ef_{2020} \leq 0.45 \text{ then } 0 \text{ else } |0.4 - ef_{2020}| + \\ &\text{if } 665 \leq Yield_{Mil} \leq 735 \text{ then } 0 \text{ else } |700 - Yield_{Mil}|/3 \end{aligned} \quad (2)$$